

SurvivalGPU

survivalGPU: scalable survival analysis in R and python

Antoine Poirot-Bourdain

CEREMADE : Université Paris Dauphine PSL, CNRS

HeKA : INRIA, INSERM, Université Paris Cité

04/02/2026

Introduction

Heka (INRIA, INSERM, Université Paris Cité)

- Anne-Sophie Jannot : public health physician
- Jean Feydy : CR INRIA
- (former) Nicolas Loche : Intern
- (former) Alexis van Straaten : Engineer
- (former) Pierre Sabatier : PhD student in public health

Ceremade (CNRS, Université Paris Dauphine PSL)

- Emmanuel Bacry : DR CNRS
- Antoine Poirot-Bourdain : PhD student in applied mathematics

Pharmacovigilance

- Adverse drug reactions are detected by the general practitioner.
- They are poorly detected.
- They are responsible for 8% of all hospitalizations.

On large-scale data from the SNDS

- Access to the social security data of 99% of the French population.
- Allows us to systematically study adverse drug reactions.
- With longitudinal survival analysis models.

A GPU powered library for scalable survival analysis

- The SNDS has millions of patients
- The WCE model is computationally intensive, especially when using bootstraps
- GPU acceleration is a game changer

In R and Python

- Survival analysis is done by physicians and epidemiologists
- They mostly use R
- We need to adapt to the community

Available functions

- Coxph
- WCE

Following the covention of reference packages

- In R : R survival and the WCE package
- In Python : sickit-learn

Statistical modelling

The hazard function

$$h(t) = \lim_{\Delta t \rightarrow 0^+} \frac{\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

The hazard function of the Cox Model

$$h(t \mid \mathbf{Z}(t)) = h_0(t) \exp \left(\sum_{s=1}^q \eta_s Z_s(t) \right)$$

Partial likelihood of the Cox model

$$L(\boldsymbol{\eta}) = \prod_{i=1}^k \frac{\exp(\boldsymbol{\eta}^\top \mathbf{X}_i(t_i))}{\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}^\top \mathbf{Z}_j(t_i))}$$

The hazard ratio

$$HR(\mathbf{Z}(t), \mathbf{Z}_*(t)) = \frac{h(t \mid \mathbf{Z}(t))}{h(t \mid \mathbf{Z}_*(t))} = \exp \left(\sum_{s=1}^q \eta_s Z_s(u - u_*) \right)$$

The hazard ratio against the baseline hazard

$$HR(\mathbf{Z}(t)) = \frac{h(t \mid \mathbf{Z}(t))}{h_0(t)} = \exp \left(\sum_{s=1}^q \eta_s Z_s(u) \right)$$

The WCE model interprets the drug exposure as a convolution

$$WCE(u) = \sum_{t=0}^u w(u-t) X(t)$$

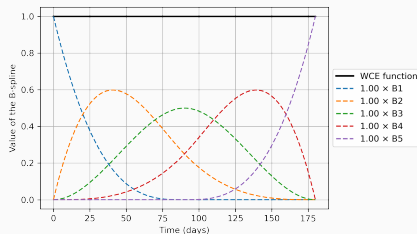
And adds it to the hazard function

$$h(u \mid \mathbf{X}(u), \mathbf{Z}(u)) = h_0(u) \exp \left(\beta \sum_{t=0}^u w(u-t) X(t) + \sum_{s=1}^q \eta_s Z_s(u) \right)$$

Modeling the WCE Function

Modeling $w(u - t)$ as a combination of cubic B-splines

$$w(u - t) = \sum_{j=1}^m \theta_j B_j(u - t)$$

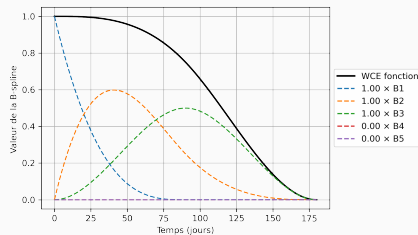


All θ parameters set to 1

Modeling the WCE Function

Modeling $w(u - t)$ as a combination of cubic B-splines

$$w(u - t) = \sum_{j=1}^m \theta_j B_j(u - t)$$

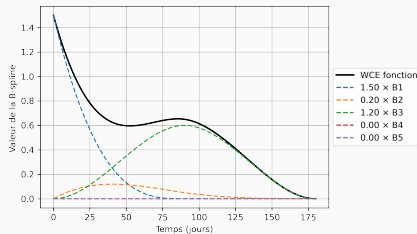


Right constraint

Modeling the WCE Function

Modeling $w(u - t)$ as a combination of cubic B-splines

$$w(u - t) = \sum_{j=1}^m \theta_j B_j(u - t)$$



Arbitrary θ parameters

We approximate the WCE function by a combination of cubic B-splines

$$w(u - t) = \sum_{j=1}^m \theta_j B_j(u - t)$$

$$h(u \mid X(u), Z(u)) = h_0(u) \exp \left(\beta \cdot \sum_{t=1}^u \sum_{j=1}^m \theta_j B_j(u - t) X(t) + \sum_{s=1}^q \eta_s Z_s(u) \right)$$

And reorganize

$$h(u \mid X(u), Z(u)) = h_0(u) \exp \left(\sum_{j=1}^m \beta \theta_j \sum_{t=1}^u B_j(u - t) X(t) + \sum_{s=1}^q \eta_s Z_s(u) \right)$$

We pose

$$\gamma_j = \beta \theta_j$$

$$D_j(u) = \sum_{t=0}^u B_j(u-t)X(t)$$

$$h(u \mid X(u), Z(u)) = h_0(u) \exp \left(\sum_{j=1}^m \beta \theta_j \sum_{t=1}^u B_j(u-t)X(t) + \sum_{s=1}^q \eta_s Z_s(u) \right)$$

And we obtain a new Cox model

$$h(u \mid X(u), Z(u)) = h_0(u) \exp \left(\sum_{j=1}^m \gamma_j D_j(u) + \sum_{s=1}^q \eta_s Z_s(u) \right)$$

- Number of interior knots (1-3)
- A eventual constraint (Left or Right)
- Cutoff (Time window where we apply the convolution)

Interprétation of the hazard Ratio

$$\text{HR} = \exp \left(\sum_{j=1}^m \hat{\gamma}_j \sum_{t=0}^u B_j(u-t) [X_1(t) - X_0(t)] \right)$$

- X_1 : one dose during all the cutoff
- X_0 : no doses during the whole cutoff
- We normalize the WCE function so its sum is equal to one during the whole cutoff

$$\text{HR} = \exp(\beta)$$

GPU acceleration



Fast computation of the convolutions even without a GPU



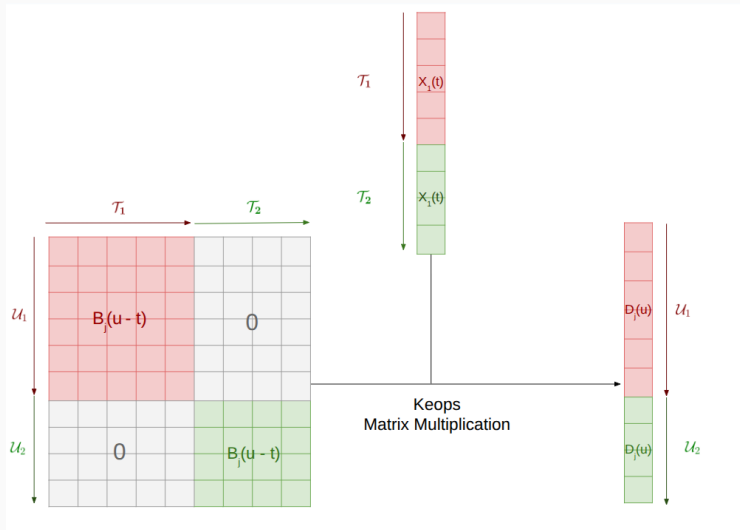
GPU-acceleration of the maximization of the likelihood

Maximization of the Cox Partial likelihood

$$L(\beta) = \prod_{i=1}^k \frac{\exp(\beta^\top \mathbf{X}_i(t_i))}{\sum_{j \in R(t_i)} \exp(\beta^\top \mathbf{X}_j(t_i))}$$

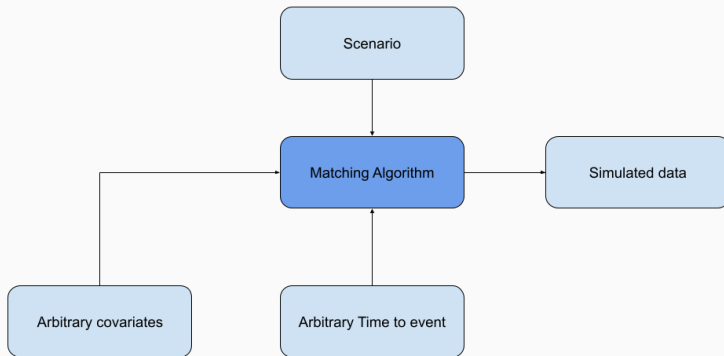
- Maximized using the Newton's method
- Can easily be done with torch
- The gradient and Hessian are automatically calculated

Computation of the convolution with pyKeops



Validadation of the WCE model

Simulation with the permutation algorithm



The simulation framework

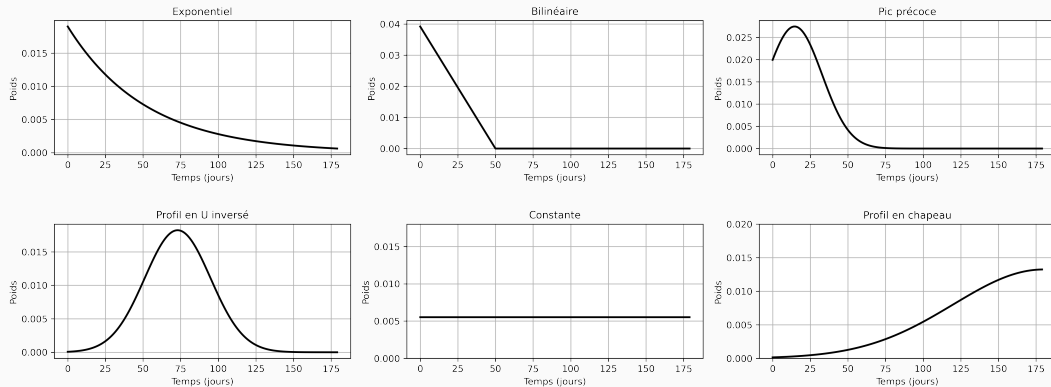
Doses

- Alternation of periods with medication and without medication
- generated by a normal distribution with mean $\log(0.5)$ and standard deviation $\log(0.8)$
- During each period with dose, random selection of a dose from: 0.5, 1, 1.5, 2, 2.5 and 3

Time-to-event data

- Generate n event times: $U[0,365]$
- Generate n censoring times: $U[0,730]$
- If the censoring time is less than the event time, then there is censoring
- Corresponds to approximately 50% censoring

The WCE scenarios



The weight functions used in the simulation experiment

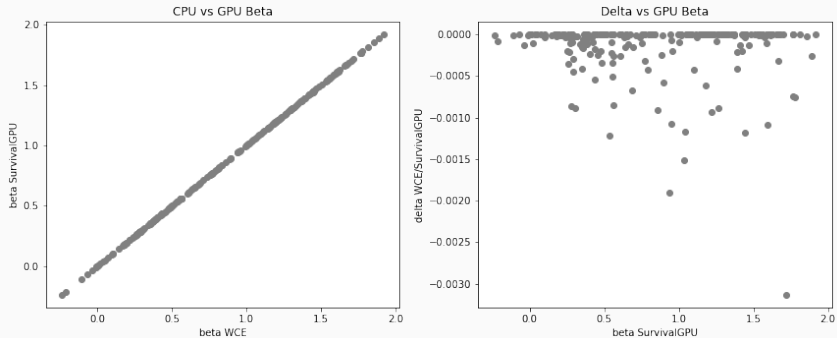
We generate 100 datasets of 500 patients for each situation

- $\beta = \log(4)$ or $\beta = \log(2)$
- Every scenario and the null hypothesis

Then for each dataset

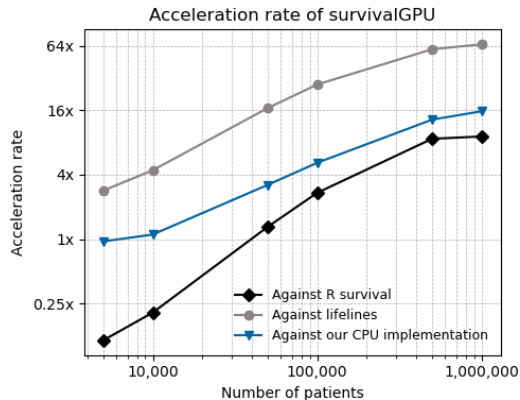
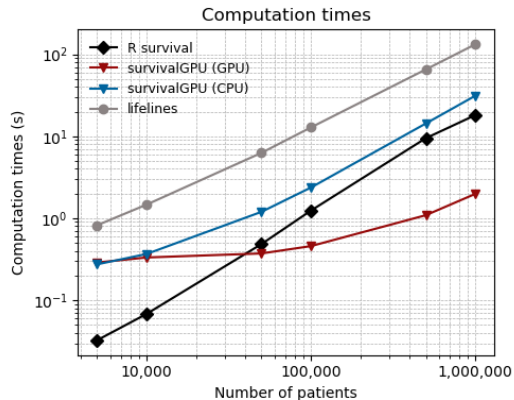
- Launch the model of the two packages (with 1,2 and 3 knots)
- Select the best nknots for each package
- Calculate the difference between the betas of the two models

Validation of the WCE model

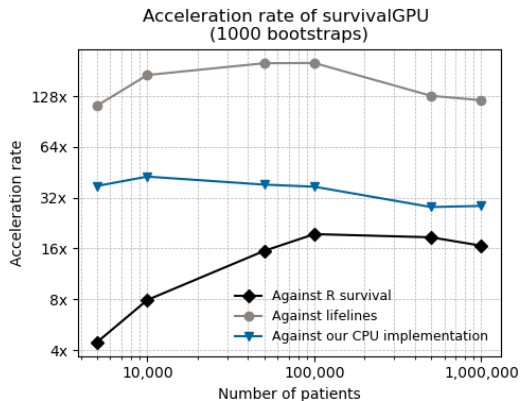
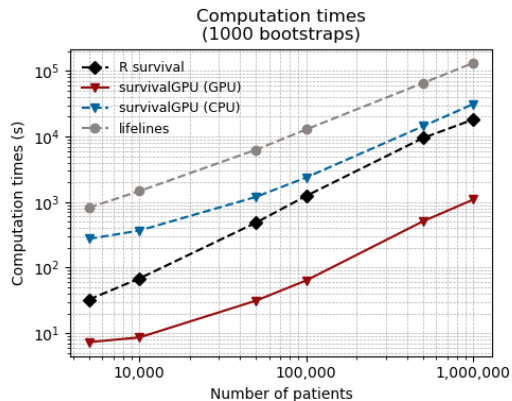


Benchmark of the of the Cox model

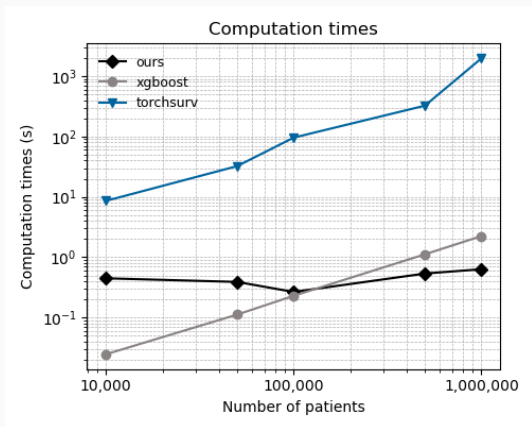
Benchmark of the Cox model



Benchmark of the Cox model with 1000 bootstraps

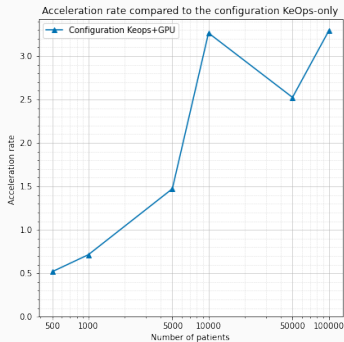
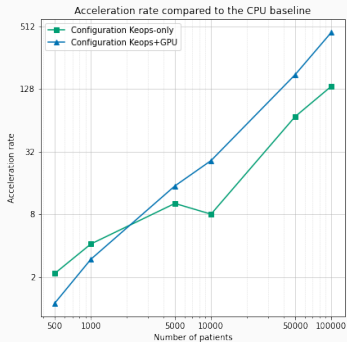
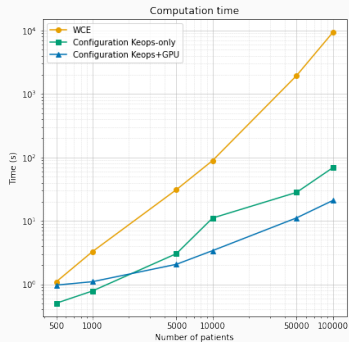


Comparison with other GPU packages

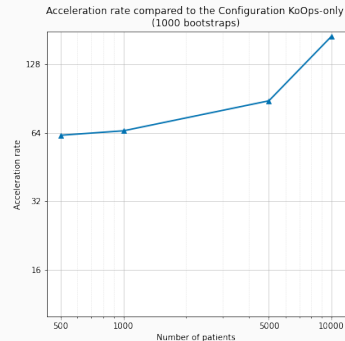
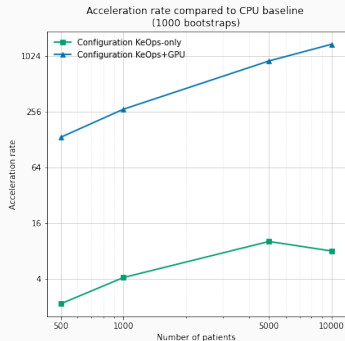
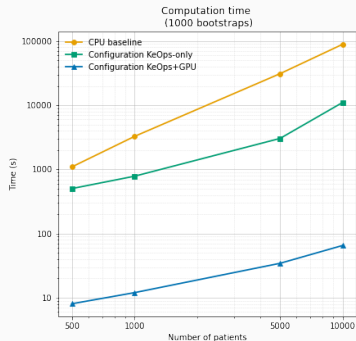


Benchmark of the of the WCE model

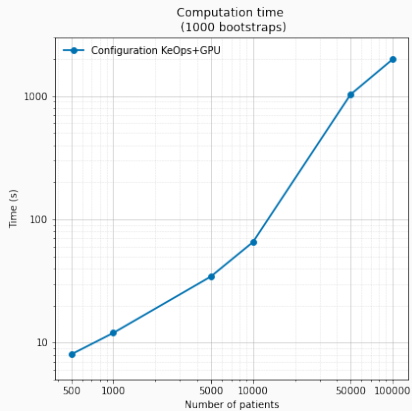
Benchmark of the WCE model



Benchmark of the WCE model with 1000 bootstraps



Benchmark of the WCE model with 1000 bootstraps



Conclusion

Validation of the package

- The Cox function of the package is validated against the survival package
- There are small differences because of the use of float32 for GPU optimization

Acceleration of the model

- The Cox is faster for large datasets of patients but the survival model was already very fast (x8 for 1,000,000 patients)
- We are of similar speed or faster than other GPU implementations
- Our package greatly accelerate bootstraps for large datasets (x20 for 1,000,000 patients and 1000 bootstraps)

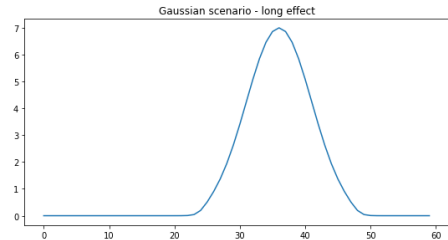
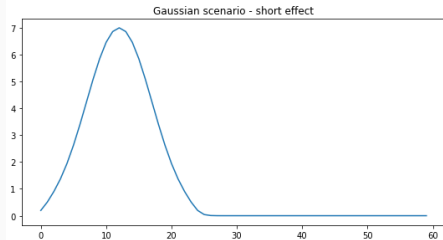
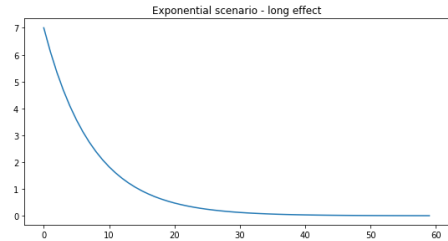
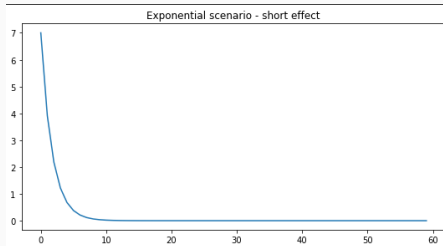
Validation of the package

- The WCE function of the package is validated against the WCE package
- There are small differences because of the use of float32 for GPU optimization

Acceleration of the model

- Our package accelerates the WCE model (x500 for 100,000 patients)
- Our package greatly accelerates bootstraps for large datasets (x5000 for 100,000 patients and 1000 bootstraps)

Limits of WCE



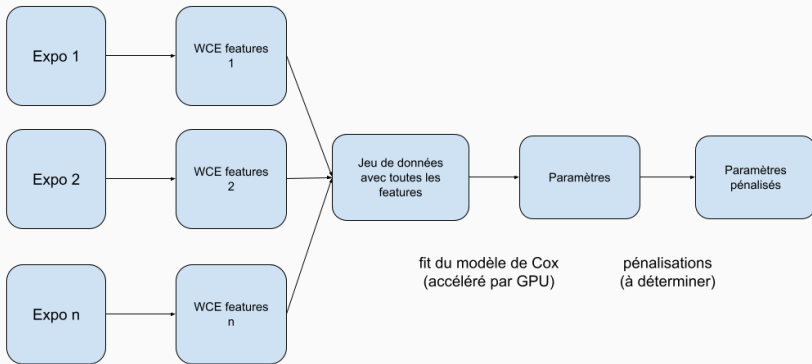
The Extended WCE model

- The WCE model can currently model the effect of only one drug
- Adverse drug reactions have often multiple causes
- With faster computations we can add more drugs to the model

Definition of the WCE function for more than one drug

$$\text{WCE}(u) = \sum_{\ell=1}^L \sum_{t=0}^u w_{\ell}(u-t) x_{\ell}(t)$$

The Extended WCE model



3 fois la fonction `wce_features`
(très rapide grâce à KeOps)